

Roll No.:

END SEMESTER EXAMINATION JUNE 2025

Name of the Program: B.Tech (ECE)

Semester: VI

Name of the Course: OOPs Using C++

Course Code: TOE-610

Time: 3 Hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main questions
- iii. Total marks in each main question are twenty.
- iv. Each questions carries 20 marks

Q1	(20Marks)	
(a)	Explain the features of Object oriented language with examples.	CO1
(b)	What do you understand by an instance variable, local variable and global variable in C++?	CO1
(c)	Write a C++ program to input any string and count number of vowels, uppercase, lowercase alphabets in a given string.	CO1
Q2		
(a)	What do you mean by Constructor in C++? Write a C++ program, which shows use of constructor in base and sub class.	CO2
(b)	What is the difference between static and non-static member in C++? Explain with the help of C++ program.	CO2
(c)	Write a C++ program to create a class Called ADemo with following members: Private Data Member: int num; Public Function Members: Parameterized constructor to initialize value of num. void evenoddCheck → This function will check whether value of data member num is even or odd. In main function create object of class ADemo and perform the above task.	CO2
Q3		
(a)	What is multiple inheritance? When does ambiguity arise in multiple inheritance? How one resolves it? What is the use of Virtual base class in C++.	CO4
(b)	Write a C++ program to overload binary + operator by using friend function.	CO3
(c)	Write a C++ program to overload binary == (equal to) operator.	CO3
Q4		
(a)	Illustrate function overloading with the help of C++ program.	CO5
(b)	Explain following with help of C++ program. (i)Virtual function (ii)Pure Virtual function	CO5 CO4

(c)	Explain multilevel and hierarchical inheritance with help of C++ program.	
Q5		CO5
(a)	Create an abstract class CapitalData with following pure virtual function: virtual void checkdata(string)=0; Create a subclass UserData which inherit abstract class CapitalData and override the function. void checkdata(String) method will check whether first character of a passing string is in uppercase or not.	CO6
(b)	What is the use of try, catch and throw keyword in C++? Explain with help of C++ program.	CO6
(c)	Write a C++ program to display largest among three numbers using function templates.	